

# Cost–Benefit Evaluation of the AI-Based E-Waste Battery Extraction System

## 1. Introduction

This proposed system utilizes AI-based CV, robotics, and object recognition technology to recognize and extract lithium-ion batteries from electronic waste. The purpose of implementing this system is to enhance efficiency, protect personnel from toxic exposure, and enhance the quality of recovered batteries.

This analysis will compare the proposed automated system with manual techniques that are currently employed for e-waste recycling.

## 2. Cost–Benefit Analysis Framework

The economic feasibility of the project can be represented using the following Cost–Benefit Analysis (CBA) equation:

**Net Benefit=(Labour Savings+Productivity Gain+Safety Improvement+Material Recovery Value)–(Hardware Cost+Development Cost+Maintenance Cost)**

This model evaluates both direct and indirect financial benefits generated by the system.

## 3. Benefits of the Proposed Computer Vision System

### 3.1 Labour Cost Reduction

Manual battery extraction requires continuous human labour, safety supervision, and protective equipment.

Manual Sorting

- Average worker wage in Australia:
- approximately AUD \$25–\$35 per hour
- One 8-hour shift:
- approximately AUD \$200–\$280 per day
- Annual labour cost per worker:
- approximately AUD \$52,000–\$70,000
- Automated CV System

The proposed robotic vision system can:

- operate continuously (24/7),

- require minimal human supervision,
- reduce repetitive manual labour,
- decrease operational downtime.
- Estimated Savings

Replacing one manual sorting station could save:

- \$55,000 to \$70,000 annually

Depending on operating hours and workforce size.

### 3.2 Increased Sorting Speed and Throughput

Human operators experience fatigue, inconsistency, and reduced concentration over long operating periods.

#### **Manual Sorting Performance**

Average processing rate:

- 10–15 items/minute
- Error rate increases during extended shifts.

#### **AI Vision System Performance**

Using:

- YOLOv8,
- segmentation workflows from Roboflow Universe,
- and edge AI deployment,

The system can process:

- 40–60 items/minute
- with consistent detection accuracy.

#### **Net Benefit**

- Approximately 4× throughput increase
- Faster processing of e-waste streams
- Reduced recycling backlog
- Higher operational productivity

### 3.3 Improved Detection Accuracy

The project uses:

- object detection,
- segmentation,
- object counting,
- and tracking pipelines.

The implemented workflow:

1. Detects batteries using YOLO-based detection.
2. Segments battery regions using mask-based segmentation.
3. Tracks objects across frames using tracking algorithms.
4. Generates labelled outputs and prediction metadata.

#### Accuracy Comparison

| Method                 | Accuracy |
|------------------------|----------|
| Manual Sorting         | 70–80%   |
| AI-Based Vision System | 90–95%   |

#### Advantages

Higher accuracy results in:

- improved material purity,
- reduced contamination,
- higher resale value of recovered black mass,
- lower downstream processing costs.

### 3.4 Safety and Risk Reduction

Battery handling is one of the most dangerous tasks in e-waste processing.

#### Key Hazards

Lithium-ion batteries may:

- overheat,
- leak chemicals,
- ignite,
- or explode due to thermal runaway.

#### Role of Computer Vision

The AI system can identify:

- swollen batteries,
- damaged battery packs,
- exposed terminals,
- overheating indicators,

- and hazardous battery conditions.

This reduces direct human exposure to:

- toxic materials,
- fire risks,
- electrical hazards.
- Financial Impact

Preventing even a single battery fire may save:

- equipment replacement costs,
- operational downtime,
- insurance penalties,
- environmental damage costs.

Potential losses avoided may exceed:

- \$100,000+
- for medium-scale recycling facilities.

## 4. System Costs

### 4.1 Hardware Costs (Capital Expenditure)

| Component                    | Estimated Cost (AUD) |
|------------------------------|----------------------|
| RGB Camera / Webcam          | \$80–\$150           |
| Intel RealSense Depth Camera | \$450–\$650          |
| NVIDIA Jetson Nano / Orin    | \$250–\$900          |
| LED Lighting System          | \$50–\$100           |
| Robotic Gripper Integration  | \$500–\$1,500        |
| Conveyor Integration         | \$400–\$1,000        |
| Miscellaneous Components     | \$200–\$500          |

Estimated Total Hardware Cost = \$1,500 – \$4,500

Depending on the deployment scale.

### 4.2 Software and Development Costs

Dataset Development

The project includes:

- custom e-waste image collection,
- annotation using CVAT,
- segmentation mask creation,
- training data preparation.

Datasets used include:

- Roboflow E-Waste Dataset
- Kaggle Recyclability Dataset
- Development Activities
- YOLOv8 model training
- segmentation model tuning
- workflow deployment
- inference optimization
- testing and validation

Estimated Development Costs

| Item                 | Estimated Cost  |
|----------------------|-----------------|
| Annotation Time      | 20–40 hours     |
| Cloud GPU Training   | \$20–\$50/month |
| Testing & Validation | \$100–\$300     |
| Model Optimization   | \$100–\$200     |

5. Comparative Evaluation

| Feature            | Manual Sorting         | AI-Based Battery Extraction System | Net Benefit           |
|--------------------|------------------------|------------------------------------|-----------------------|
| Processing Speed   | 10–15 items/min        | 40–60 items/min                    | 4× Faster             |
| Detection Accuracy | 70–80%                 | 90–95%                             | Higher Quality        |
| Operating Hours    | Limited by shifts      | 24/7 Operation                     | Continuous Processing |
| Labour Cost        | Recurring hourly wages | Low maintenance cost               | Major Savings         |
| Worker Safety      | High Risk              | Minimal Human Exposure             | Critical Improvement  |
| Scalability        | Difficult              | Easily Scalable                    | Industrial Expansion  |
| Data Collection    | Manual records         | Automatic digital logging          | Better Monitoring     |

6. Scalability and Industrial Impact

The proposed system is highly scalable and suitable for:

- recycling centres,
- smart waste facilities,
- robotic disassembly stations,
- Industry 4.0 manufacturing environments.

Future improvements may include:

- real-time defect classification,
- thermal camera integration,
- robotic arm automation,
- multi-class hazardous waste detection,
- cloud-based monitoring systems.

The integration of AI and robotics into e-waste recycling supports sustainable waste management and contributes to safer and more efficient circular economy practices.

## 7. Conclusion

The AI-based e-waste battery extraction system offers significant economic and operational benefits over manual sorting methods. Although the initial setup cost is higher, the system quickly recovers its investment through labor savings, faster processing, improved detection accuracy, and safer working conditions.

The greatest advantage of this suggested approach is not only the improvement of productivity, but also minimizing human contact with risky tasks involving lithium-ion batteries. Thus, utilizing Computer Vision and robotics technologies can become an effective solution in modern e-waste facilities.

## References

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